LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 8: Limitations and Challenges
Lecture	Lecture 8.4: Ethical and Safety Concerns

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Identify the primary ethical and safety concerns associated with generative AI deployment.
- Explain the risks of AI-enabled misinformation and disinformation at scale.
- Describe the copyright, intellectual property, and privacy concerns that arise from AI training and deployment.
- Apply an ethical risk assessment framework to a professional AI deployment scenario.
- Identify the organisational governance mechanisms necessary for responsible AI use.

Introduction

Generative AI's capabilities—scale, speed, and quality of content generation—are precisely the properties that make its ethical risks so significant. The same system that can produce a thousand helpful customer service responses in an hour can also produce a thousand pieces of convincing misinformation. The same image generation capability that enables rapid prototyping of design concepts can enable sophisticated deepfake creation. Ethical responsibility in AI is not separate from practical AI use—it is inseparable from it, because the harms are real and the practitioners who deploy these systems are in a position to prevent or perpetuate them.

The ethical landscape of generative AI is still evolving rapidly, and the regulatory environment is developing unevenly across jurisdictions. Practitioners who wait for comprehensive regulation before engaging with AI ethics will find themselves behind. Those who build ethical considerations into deployment practices now—through governance structures, impact assessments, transparency measures, and ongoing monitoring—will be better positioned to comply with emerging regulations and to avoid the reputational, legal, and social harms that poorly governed AI deployments produce.

This guide addresses the four primary ethical domains covered in the lecture—misinformation, intellectual property, privacy, and environmental impact—and provides practical governance tools for responsible AI deployment at the organisational level.

Core Concepts Explained

Misinformation, Disinformation, and Synthetic Media Risks

Definition: Generative AI enables the production of convincing false text, images, audio, and video at unprecedented scale and speed. Misinformation (false information spread unintentionally) and disinformation (false information spread deliberately) are both enabled and amplified by these capabilities.

Business relevance: Organisations face both the risk of being targeted by AI-generated disinformation (false content about their products, executives, or actions) and the risk of inadvertently deploying systems that produce or amplify false information. Both risks require active management.

AI Practitioner relevance: AI practitioners deploying content generation tools must implement safeguards against the production of false or misleading content. These safeguards include fact-checking workflows, content verification requirements, use-case restrictions, and monitoring for misuse of deployed systems.

Real-world example: During the 2024 election cycle in multiple countries, AI-generated deepfakes of political candidates were circulated on social media platforms—including a widely circulated AI-generated audio clip falsely depicting a UK political leader making offensive remarks. Platforms including Meta and YouTube updated their policies to require disclosure of AI-generated content in political advertising, but enforcement remains challenging at scale.

Risks or limitations: The 'liar's dividend'—the ability to deny authentic content by claiming it is AI-generated—may ultimately be as harmful as the production of false content. As synthetic media becomes prevalent, the credibility of authentic video and audio evidence may be undermined regardless of whether AI is actually used to create a specific piece of content.

Copyright, Intellectual Property, and Creator Rights

Definition: Generative AI models are trained on large corpora of copyrighted works—books, articles, art, code, music—often without explicit permission from rights holders. The legal status of this training practice, and of the outputs generated from these models, remains contested in courts and legislatures globally.

Business relevance: Organisations using AI-generated content face legal uncertainty about copyright ownership and potential liability for outputs that closely resemble copyrighted works. Legal risk exposure varies by jurisdiction, use case, and the degree to which outputs can be traced to specific training inputs.

AI Practitioner relevance: AI practitioners should advise their organisations on the IP risks of AI-generated content—particularly for externally published content, commercial products, and content in domains with active litigation (art, music, code, writing). Engaging legal counsel with AI expertise is advisable for any high-stakes commercial use of AI-generated content.

Real-world example: In 2023, a coalition of authors including prominent novelists filed suit against OpenAI and other AI companies, alleging that their copyrighted books were used to train language models without permission or compensation. Separately, Getty Images filed suit against Stability AI for training image generation models on Getty's licensed photo library. Both cases remained unresolved as of 2025, with significant implications for the training data practices of all generative AI companies.

Risks or limitations: The legal uncertainty is genuine—courts in different jurisdictions are reaching different conclusions, and the outcome of major pending cases will significantly shape the legal landscape. Organisations with significant IP exposure in AI-generated content should track these cases and prepare to adjust practices as legal clarity emerges.

Privacy and Data Governance

Definition: Generative AI raises privacy concerns at two stages: training (models may have ingested personal information from training data) and deployment (user interactions with AI systems may be used for further training or retained by providers). Both stages create data governance obligations for organisations deploying these systems.

Business relevance: Organisations operating under GDPR, CCPA, or equivalent privacy regulations must assess whether their AI deployments comply with data protection requirements. This includes assessing what personal data users share with AI systems, how that data is retained and processed by the AI provider, and whether appropriate consents and safeguards are in place.

AI Practitioner relevance: AI practitioners must ensure that AI deployments are covered by data processing agreements with providers and that the types of data users can share with AI systems are consistent with data protection obligations. Sensitive categories of personal data should not be shared with AI systems without specific legal basis and safeguards.

Real-world example: In 2023, Samsung engineers were found to have shared proprietary source code with ChatGPT while using it to assist with debugging, inadvertently exposing trade secrets to a third-party system. Samsung subsequently banned ChatGPT for internal use pending the development of an in-house AI solution. The incident illustrated the data governance risks of employees using public AI tools with sensitive internal information.

Risks or limitations: Many AI providers have updated their terms of service to allow opt-out from training data use; however, retention, processing, and access policies vary significantly. Practitioners should review current provider terms carefully and not assume that enterprise subscriptions provide complete data isolation without verifying the specific terms.

Environmental Impact of Large-Scale AI

Definition: Training and running large AI models consumes significant computational resources and associated energy. Training frontier models requires large-scale GPU clusters running for weeks to months; inference (generating outputs from deployed models) at scale also represents substantial ongoing energy consumption.

Business relevance: Organisations with sustainability commitments and net-zero targets should account for AI-related energy consumption in their carbon accounting. AI vendors are increasingly publishing energy consumption and carbon intensity data, enabling more informed procurement decisions for sustainability-conscious organisations.

AI Practitioner relevance: AI practitioners can contribute to sustainability goals through task-model matching (using smaller models where adequate), optimising prompt efficiency to reduce unnecessary token generation, and factoring energy costs into AI tool procurement criteria alongside capability and price.

Real-world example: Google's 2024 environmental report disclosed that AI-related energy consumption was a significant contributor to the company's overall energy usage and presented a challenge to its net-zero commitments. Microsoft made similar disclosures, noting that AI infrastructure expansion had increased its carbon footprint despite investments in renewable energy. These disclosures prompted enterprise customers to begin including AI energy consumption questions in vendor due diligence.

Risks or limitations: As AI capabilities improve and usage scales, the aggregate environmental impact of AI infrastructure is likely to increase significantly. Organisations that ignore this dimension now may face regulatory reporting obligations, stakeholder pressure, and reputational risks as AI sustainability becomes a more prominent public and regulatory concern.

Practical Applications

The following scenarios illustrate ethical governance mechanisms applied in professional AI deployments.

Responsible Content Generation Policy

A media company deploying AI for content generation establishes a responsible use policy covering four areas: (1) Accuracy—all AI-generated factual claims must be verified against primary sources before publication. (2)

Disclosure—all published content with substantial AI contribution carries a disclosure statement. (3) IP compliance—AI-generated content involving specific artistic styles or character names is reviewed by legal before publication. (4) Prohibited uses—AI generation of content about real individuals in fictional scenarios, synthetic quotes attributed to real people, and political disinformation are explicitly prohibited. The policy is embedded in the content workflow as a mandatory review step.

Data Governance for AI Tools

A professional services firm establishes a tiered data governance policy for AI tool use: (1) Public tools (ChatGPT, Gemini, Claude.ai)—permitted for general research and drafting with non-confidential information only; personal data and client information prohibited. (2) Enterprise-licensed tools (Microsoft Copilot)—permitted for internal work including client project information, subject to data processing agreements with providers. (3) Custom in-house tools—permitted for all data types within applicable regulations. Employees receive annual training on the policy; violations are treated as data protection incidents.

AI Ethics Impact Assessment

Before deploying a new AI use case, a financial services organisation completes a standardised AI Ethics Impact Assessment covering: affected stakeholders and potential harms; fairness and bias risk; transparency and explainability requirements; privacy and data governance compliance; environmental impact; and escalation criteria for high-risk deployments requiring board-level approval. The assessment is completed by the product team, reviewed by the AI governance function, and signed off by a senior executive for any deployment rated as high-risk. Completed assessments are maintained as evidence of due diligence.

Best Practices

- **Establish an AI use policy before deploying AI tools at scale.** A clear policy defining permitted and prohibited uses, data handling requirements, disclosure obligations, and escalation criteria is the foundation of responsible AI governance. The absence of a policy does not prevent misuse—it just means misuse occurs without oversight.
- **Require AI disclosure in externally facing content.** Consumers, clients, and regulators are increasingly expecting disclosure of AI involvement in content generation. Establishing proactive disclosure practices now builds trust and prepares for mandatory disclosure requirements that are emerging in multiple jurisdictions.
- **Conduct AI Ethics Impact Assessments for new deployments.** Formalised impact assessment—covering stakeholder harms, fairness, privacy, transparency, and environmental impact—ensures that ethical considerations are addressed systematically before deployment, not reactively after incidents.
- **Engage legal counsel for IP-sensitive AI applications.** Copyright uncertainty around AI-generated content is a real legal risk, not a theoretical one. For any commercial application of AI-generated content, legal review by counsel with AI IP expertise is advisable.
- **Include AI energy consumption in sustainability reporting.** AI-related energy use is a growing component of organisational carbon footprints. Including it in sustainability reporting—and factoring it into AI tool procurement—demonstrates genuine environmental accountability.

Common Mistakes and Misconceptions

Misconception: Assuming AI-generated content is copyright-free and risk-free to publish commercially.

Correct approach: The copyright status of AI-generated content is legally contested and varies by jurisdiction. AI outputs may also closely resemble copyrighted training inputs. Legal review is advisable for any commercial publication of AI-generated content, particularly in domains with active IP litigation.

Misconception: Allowing employees to share sensitive client or proprietary data with public AI tools without governance.

Correct approach: Public AI tools typically process and may retain input data; sharing confidential or personal data with them may violate contractual obligations, privacy regulations, and trade secret protections. A tiered data governance policy specifying which data can be shared with which types of AI tools is essential.

Misconception: Treating AI ethics as a one-time compliance exercise rather than an ongoing practice.

Correct approach: AI systems evolve, use cases expand, and the ethical risk landscape changes. Ethical governance must be continuous: regular policy reviews, ongoing monitoring of deployed systems, and updating impact assessments when use cases change.

Misconception: Ignoring environmental considerations in AI tool selection and workflow design.

Correct approach: AI energy consumption is measurable and material. Practitioners can reduce it through task-model matching, prompt efficiency, and sustainability-informed procurement. Ignoring this dimension is increasingly inconsistent with organisational sustainability commitments.

Assessment Framework: AI Ethics Impact Assessment Template

Complete this assessment before deploying any new AI use case. High-risk ratings require escalation to senior leadership before deployment approval.

Ethical Dimension	Assessment Questions	Risk Rating (L/M/H)	Mitigation Plan
Harm to stakeholders	Who is affected? What potential harms exist?		
Fairness and bias	Could outputs disadvantage protected groups?		
Transparency	Will stakeholders know AI is involved?		
Privacy and data	What personal data is involved? Is consent in place?		
IP and copyright	Does content risk infringing third-party rights?		
Environmental impact	What is the projected energy consumption?		
Misuse potential	Could the system be misused to cause harm?		

How to use this: *Any dimension rated High requires a documented mitigation plan and senior leadership sign-off before deployment. Completed assessments should be retained as evidence of due diligence. Review and update when the use case changes significantly.*

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. An employee uses a public AI tool to help draft a client proposal and includes specific client financial data in the prompt. What is the primary risk?

A. The AI output will be lower quality due to the sensitive data included.

B. The client financial data may be processed and potentially retained by the public AI provider, violating confidentiality obligations.

C. The AI tool will refuse to process financial data.

D. The proposal will not be copyright-protected because AI was involved.

Answer: B. The client financial data may be processed and potentially retained by the public AI provider, violating confidentiality obligations.

Why: *Public AI tools process input data according to provider terms, which often allow retention for training or quality improvement. Sharing client confidential data with a public AI tool may breach the client confidentiality agreement and applicable data protection regulations.*

2. A technology company uses AI to generate stock imagery for its marketing website. What copyright-related step is most important before publishing?

A. None—AI-generated images are automatically copyright-free.

B. Legal review to assess IP risks, given contested copyright status of AI-generated content and potential resemblance to training images.

C. Confirm the AI tool's terms of service allow commercial use.

D. Add a copyright notice attributing the images to the AI tool.

Answer: B. Legal review to assess IP risks, given contested copyright status of AI-generated content and potential resemblance to training images.

Why: *The copyright status of AI-generated content is legally contested; AI-generated images may also resemble copyrighted training inputs. Legal review is advisable for commercial publication, particularly given active litigation in this area.*

3. An organisation's sustainability team notices that AI tool usage has increased significantly over the past year. What is the most appropriate next step?

A. Reduce AI usage across the organisation to meet sustainability targets.

B. Quantify AI-related energy consumption and include it in sustainability reporting, then explore efficiency optimisations such as task-model matching.

C. Offset AI energy consumption by purchasing renewable energy credits.

D. Request that AI vendors provide zero-carbon guarantees before continuing use.

Answer: B. Quantify AI-related energy consumption and include it in sustainability reporting, then explore efficiency optimisations such as task-model matching.

Why: *Accurate measurement and reporting is the foundation of responsible environmental governance; once quantified, efficiency optimisations like task-model matching can reduce consumption without eliminating AI benefit.*

Reflection Questions

1. Does your organisation have an AI use policy? If yes, does it cover data governance, IP, disclosure, and prohibited uses? If not, what would the highest-priority provisions be?

2. Think of a recent AI deployment in your organisation. Was an ethics impact assessment conducted? What ethical dimensions, if any, were not considered?
3. How would you handle a situation where you discovered that colleagues were sharing client confidential information with a public AI tool? What steps would you take?
4. As AI-generated deepfakes and synthetic media become more prevalent, how should your organisation prepare—both to defend against AI-generated disinformation targeting you, and to govern your own use of synthetic media?
5. How do you personally balance the efficiency benefits of AI tools with the ethical considerations they raise? Where do you draw your own lines?

Key Takeaways

- Generative AI's scale and speed amplify both its benefits and its ethical risks; misinformation, IP, privacy, and environmental harms are not hypothetical—they are documented and growing.
- AI-enabled misinformation and disinformation operate at scale that human content moderation cannot match; prevention through use-case restrictions, disclosure, and fact-checking workflows is more effective than post-hoc correction.
- The copyright status of AI training data and AI-generated outputs remains legally contested; organisations with commercial AI content exposure should engage legal counsel and monitor case developments.
- Data governance for AI tools is a real compliance obligation: sharing personal or confidential data with public AI tools may violate privacy regulations, contractual obligations, and trade secret protections.
- AI ethics is an ongoing practice, not a one-time compliance exercise; policies, impact assessments, and monitoring must be maintained and updated as systems evolve and use cases expand.
- Environmental impact of AI is measurable and material; task-model matching, prompt efficiency, and sustainability-informed procurement are practical levers for reducing AI's carbon footprint.
- Ethical AI governance requires organisational structures—policies, impact assessments, senior leadership accountability, and incident response processes—not just individual practitioner awareness.

Recommended Next Actions

6. Review your organisation's AI use policy (or identify the absence of one) and assess whether it covers data governance, IP risk, disclosure obligations, and prohibited uses.
7. Complete the AI Ethics Impact Assessment for one current AI deployment. Identify any High-rated dimensions and design a mitigation plan.
8. Identify colleagues who may be sharing sensitive data with public AI tools without awareness of the risk. Design a targeted communication or training to address this gap.
9. Research the AI provisions of GDPR, the EU AI Act, or equivalent regulation in your jurisdiction. Assess which of your AI use cases fall within the scope of those provisions.